

Endress & Hauser Promag 800 Email driver

Introduction

This driver is designed to provide an interface for Promag 800 flow meters.

The data is sent in the form of an email containing a csv attachment.

Once the email is received it is decoded to expose the values contained in the attachment and body of the mail.

Inbox management is provided through a backup box and an unknown/todo box, where email messages can be moved as appropriate.

Messages older than a configured amount of days may be set for auto deletion.

A separate Adroit device has to be created for each flow meter with the station ID being set equal to the serial number of the flow meter as per the CSV file that get emailed.

There is a resource id allocated automatically on configuration, new devices will attempt to find a compatible configuration (those that share an inbox for example) any others will create a new resource, selecting a resource in the dialogue will load the current configuration for that box, if the device is saved (On OK) any changes will be save with an appropriate warning. Resource ID can be deleted with appropriate warnings if it might affect other devices.

Offline devices and unknown (unconfigured) devices will have their emails moved to the UNKNOWN box if the MOVE radio is applied. Else if DELETE was selected unconfigured emails will be deleted. And offline device emails will be left in the main INBOX.

Configuration Dialog

EHPROMAG E-mail Client Protocol Driver : PROMAG01

Mail server settings (*Resource key)

Resource: Del

*Mail Server:

*Mail Port:

*Login Name:

Login Password:

*Mailbox Type

☒ IMAP

☐ POP

☒ *Server uses SSL.

☐ Detailed Debugging on Datascope.

Incoming mailbox configuration

Incoming Mailbox:

Poll Period (ms):

☐ Move used messages to BACKUP mailbox

☒ Delete used messages

☐ Move unknown messages to UNKNOWN mailbox

☒ Delete unknown messages

Local Device Settings

Station Id:

Model:

☒ Device contains Units.

Backup mailbox configuration

Backup Mailbox:

Delete messages older than days (0 - disabled)

Unknown mailbox configuration

Unknown Mailbox:

Delete messages older than days (0 - disabled)

OK Cancel

Mail server settings (*Resource key)

(the parameters identified by the * define a resource common to all devices that share those parameters, any change to any other parameter in this group will be changed for all devices using this mailbox.)

***Mailbox Type** – Choose IMAP or POP depending on your server type. (pop has been disabled for the time being)

***Mail Server** – The name of the mail server

***Mail Port** – 143 for IMAP , 110 for POP. (Some Servers use non-default ports)

***Server uses SSL** – IF the mail server requires this, usually IMAP port will then need 993.

***Login Name** – User name on the server.

(the remaining parameters can be considered as sub-parameters (these will not result in a new resource being generated if different from any other resource) and if a change is requested on these the whole resource will be updated, and thus affect all devices using that resource)

Login Password – User password.

Detailed debugging to datascope – Additional processing information will be sent to the datascope.

Incoming mailbox configuration

Incoming Mailbox Inbox name – this is where all new emails should arrive first.

Mailbox Poll Period (ms) – Mailbox will be checked for new mail at this interval.

Move used messages to BACKUP mailbox - Radio – If a mail is parsed and accepted and considered complete, checking this radio button will tell the driver to move that mail to the used or BACKUP mailbox, where further processing could take place such as remain there for x amount of days before being discarded.(see further down)

Delete used messages - Radio – If a mail is parsed and accepted and considered complete, checking this radio button will tell the driver to delete the mail immediately.

Move unknown messages to UNKNOWN mailbox - Radio – If a mail is parsed and found to be junk or for an offline device and NOT considered complete, checking this radio button will tell the driver to move that mail to the todo or UNKNOWN mailbox, where further processing could take place such as remain there for x amount of days before being discarded.(see further down)

Delete unknown messages - Radio – If a mail is parsed and found to be junk or for an offline device and NOT considered complete, checking this radio button will tell the driver to delete the mail immediately.

Backup mailbox configuration

Backup Mailbox – Backup name – this is a sub or secondary mailbox where completed emails can be stored for a period of time.

Delete messages older than x days – Any email in this box older than this number of days will be deleted during driver maintenance tasks.

Unknown mailbox configuration

Unknown Mailbox – Unknown name – this is a sub or secondary mailbox where incompleated emails can be stored for a period of time. Junk/spam and or unconfigured or offline devices will also go here.

Delete messages older than x days – Any email in this box older than this number of days will be deleted during driver maintenance tasks.

Unknown mailbox configuration

(these setting are stored per device and will not affect other devices using the same resource id.)

Station ID – This is the identifier code sent in the subject line of the email. (for example : Data from ML250 29902 , the id is 29902.

Model – Select the Model of the Promag 800 flometer. Currently only Promag 800 is available.

Device contains units – The Promag 800 can report data with unit fields.

Supported addresses

A CSV row constitutes 1 record:

1,2010/02/03,00:00:00,<Unit 1>,<Value1>,<Unit 2>,<Value2>...<Unit

21>,<Value21>

Data	Result Representation	Adroit Integer	Adroit Real	Adroit String
Unit 1	Unit for Positive Total			U1
Value 1	Value of Positive Total	V1	V1	
Unit 2	Unit of Negative Total			U2
Value 2	Value of Negative Total	V2	V2	
Unit 3	Unit of Flow Rate			U3
Value 3	Value of Flow Rate	V3	V3	
Unit 4	Unit of Flow Rate %			U4
Value 4	Value of Flow Rate %	V4	V4	
Unit 5	Unit of SPARE			U5
Value 5	Value of SPARE	V5	V5	
Unit 6	Unit of SPARE			U6
Value 6	Value of SPARE	V6	V6	
Unit 7	Unit of Positive Partial Total			U7
Value 7	Value of Positive Partial Total	V7	V7	
Unit 8	Unit of Negative Partial Total			U8
Value 8	Value of Negative Partial Total	V8	V8	
Unit 9	Unit of Total Net Value			U9
Value 9	Value of Total Net Value	V9	V9	
Unit 10	Unit of Partial Net Value			U10
Value 10	Value of Partial net Value	V10	V10	
Unit 11	Unit of SPARE			U11
Value 11	Value of SPARE	V11	V11	
Unit 12	Unit of SPARE			U12
Value 12	Value of SPARE	V12	V12	
Unit 13	Unit of SPARE			U13
Value 13	Value of SPARE	V13	V13	
Unit 13	Unit of SPARE			U14
Value 14	Value of SPARE	V14	V14	
Unit 15	Unit of SPARE			U15
Value 15	Value of SPARE	V15	V15	
Unit 16	Unit of Cycles Per Hour			U16
Value 16	Value of Cycles Per Hour	V16	V16	
Unit 17	Unit of Battery 1 Status			U17
Value 17	Value of Battery 1 Status	V17	V17	
Unit 18	Unit of Battery 2 Status			U18
Value 18	Value of Battery 2 Status	V18	V18	
Unit 19	Unit of Battery 3 Status			U19
Value 19	Value of Battery 3 Status	V19	V19	
Unit 20	Unit of Antenna Signal			U20
Value 20	Value of Antenna Signal	V20	V20	
Unit 21	Unit of Board Temperature			U21
Value 21	Value of Board Temperature	V21	V21	

Application example

1. You have a Promag 800 flow meter from which you need to extract the following data:
 - a. The positive volume counter
 - b. The negative volume counter
 - c. The Flow rate
2. You are also required to extract the units of these values from the csv file.
3. In addition to this you need to determine the signal strength of the antenna.

In order to accomplish this you will need to create 4 analog agents and 4 string agents.

You will scan the value of the positive volume counter (analog) from the “V1” address of the device

You will scan the unit of the positive volume counter (string) from the “U1” address of the device.

You will scan the value of the negative volume counter (analog) from the “V2” address of the device

You will scan the unit of the negative volume counter (string) from the “U2” address of the device.

You will scan the value of the flow rate (analog) from the “V3” address of the device

You will scan the unit of the flow rate (string) from the “U3” address of the device.

You will scan the value of the antenna signal strength from the “V20” address of the device.

You will scan the unit of the antenna signal strength from the “U20” address of the device.

Fault finding

1. Ensure the connection settings to your mailbox is correct.
2. Ensure the serial number of the Promag 800 flow meter matches the Station ID as set up in the driver configuration.
3. Ensure the emails do have attachments.
4. Ensure that the data attachment is before the event attachment.
5. Ensure that there is no invalid data in the body of the email, or in the attachment.
6. Do not move emails between folders in excel. A Copy is required for the driver to operate to specification.

Revision information

Date:	Changes:
16-Apr-2015	First Released Document.

Appendix A – Datascope examples

```

ASCII DATASCOPE for 63088 on Port: 143
000000 : U3: 1/s : U3: 0.000000 : U4: bar [r][n] : U4: 12.898000 :
Line 16 : 2012/12/05 08:00:00:00! U1: m3 : U1: 13179.900000 : U2: m3 : U2: -15.0
000000 : U3: 1/s : U3: 0.000000 : U4: bar [r][n] : U4: 12.910000 :
: B1: 99.000000 : B2: 0.000000 : B3: 99.000000 : U8: 10.000000 : U9: 57.000000
Moved message [4] to Backup Mailbox

[5.0] Retrieve info : text/plain,quoted-printable []
[5.0] Body found
[5.1] Retrieve info : text/plain,base64 [SD063088_2012_12_04_16_00_25.CSV]
[5.1] Id determined as : 063088
[5.1] Attachment [SD063088_2012_12_04_16_00_25.CSV] found
Retrieving email completed

Starting to Parse Message
Done Parsing Message
Examine attachment for device: 63088 RealJob: 0 [22544536]
Updated (16) records for device: 63088 RealJob: 0 [22544536] details follow ==>

Line 17 : 2012/12/04 08:30:01:00! U1: : U1: 13179.500000 : U2: m3 : U2: -15.000
000 : U3: 1/s : U3: 0.031000 : U4: bar [r][n] : U4: 12.672000 :
Line 18 : 2012/12/04 09:00:03:00! U1: m3 : U1: 13179.500000 : U2: m3 : U2: -15.0
00000 : U3: 1/s : U3: 0.000000 : U4: bar [r][n] : U4: 12.861000 :
Line 19 : 2012/12/04 09:30:02:00! U1: m3 : U1: 13179.500000 : U2: m3 : U2: -15.0
00000 : U3: 1/s : U3: 0.000000 : U4: bar [r][n] : U4: 11.725000 :
Line 20 : 2012/12/04 10:00:00:00! U1: m3 : U1: 13179.500000 : U2: m3 : U2: -15.0
00000 : U3: 1/s : U3: 0.020000 : U4: bar [r][n] : U4: 12.361000 :
Line 21 : 2012/12/04 10:30:03:00! U1: m3 : U1: 13179.500000 : U2: m3 : U2: -15.0
00000 : U3: 1/s : U3: 0.000000 : U4: bar [r][n] : U4: 12.842000 :
Line 22 : 2012/12/04 11:00:01:00! U1: m3 : U1: 13179.500000 : U2: m3 : U2: -15.0
00000 : U3: 1/s : U3: 0.000000 : U4: bar [r][n] : U4: 12.154000 :
Line 23 : 2012/12/04 11:30:02:00! U1: m3 : U1: 13179.600000 : U2: m3 : U2: -15.0
00000 : U3: 1/s : U3: 0.014000 : U4: bar [r][n] : U4: 12.028000 :
Line 24 : 2012/12/04 12:00:01:00! U1: m3 : U1: 13179.600000 : U2: m3 : U2: -15.0
00000 : U3: 1/s : U3: 0.000000 : U4: bar [r][n] : U4: 12.841000 :
Line 25 : 2012/12/04 12:30:02:00! U1: m3 : U1: 13179.600000 : U2: m3 : U2: -15.0
00000 : U3: 1/s : U3: 0.000000 : U4: bar [r][n] : U4: 12.462000 :
Line 26 : 2012/12/04 13:00:01:00! U1: m3 : U1: 13179.600000 : U2: m3 : U2: -15.0
00000 : U3: 1/s : U3: 0.007000 : U4: bar [r][n] : U4: 11.483000 :
Line 27 : 2012/12/04 13:30:01:00! U1: m3 : U1: 13179.600000 : U2: m3 : U2: -15.0
00000 : U3: 1/s : U3: 0.030000 : U4: bar [r][n] : U4: 12.853000 :
Line 28 : 2012/12/04 14:00:00:00! U1: m3 : U1: 13179.600000 : U2: m3 : U2: -15.0
00000 : U3: 1/s : U3: 0.000000 : U4: bar [r][n] : U4: 12.844000 :
Line 29 : 2012/12/04 14:30:04:00! U1: m3 : U1: 13179.600000 : U2: m3 : U2: -15.0
00000 : U3: 1/s : U3: 0.000000 : U4: bar [r][n] : U4: 11.817000 :
Line 30 : 2012/12/04 15:00:04:00! U1: m3 : U1: 13179.700000 : U2: m3 : U2: -15.0
00000 : U3: 1/s : U3: 0.045000 : U4: bar [r][n] : U4: 12.444000 :
Line 31 : 2012/12/04 15:30:02:00! U1: m3 : U1: 13179.700000 : U2: m3 : U2: -15.0
00000 : U3: 1/s : U3: 0.000000 : U4: bar [r][n] : U4: 12.849000 :
Line 32 : 2012/12/04 16:00:00:00! U1: m3 : U1: 13179.700000 : U2: m3 : U2: -15.0
00000 : U3: 1/s : U3: 0.000000 : U4: bar [r][n] : U4: 12.846000 :
: B1: 99.000000 : B2: 0.000000 : B3: 99.000000 : U8: 10.000000 : U9: 48.000000
Moved message [5] to Backup Mailbox

Housekeeping - nothing to do on BACKUPBOX
Housekeeping - nothing to do on UNKNOWNBOX
Connected Successfully!IMAP Mailbox [INBOX] Empty...
Housekeeping - nothing to do on BACKUPBOX
Housekeeping - nothing to do on UNKNOWNBOX

```